

Expt 5	Design of High Pass Filter using Windowing Techniques
Date:	5.1 Generate the coefficients of an FIR High-Pass filter using Rectangular windowing techniques.

CODE:-

```

clc; clear; close all;

%% --- Input Parameters ---

fp = 1000; % High-pass cutoff frequency (Hz)
fs = 8000; % Sampling frequency (Hz)
N = 50; % Filter order

wc = fp/(fs/2); % Normalized cutoff (0–1)

%% --- 1. FIR High Pass using Rectangular Window ---
b_rect = fir1(N, wc, 'high', rectwin(N+1));
disp('--- High-Pass Coefficients (Rectangular Window) ---');
disp(b_rect);

%% --- 2. FIR High Pass using Hamming Window ---
b_hamm = fir1(N, wc, 'high', hamming(N+1));
disp('--- High-Pass Coefficients (Hamming Window) ---');
disp(b_hamm);

%% --- 3. FIR High Pass using Hanning Window ---
b_hann = fir1(N, wc, 'high', hanning(N+1));
disp('--- High-Pass Coefficients (Hanning Window) ---');
disp(b_hann);

%% --- Frequency Response Plots ---

figure;
freqz(b_rect,1);

```

```
title('High-pass FIR using Rectangular Window');  
figure;  
freqz(b_hamm,1);  
title('High-pass FIR using Hamming Window');  
figure;  
freqz(b_hann,1);  
title('High-pass FIR using Hanning Window');
```

OUTPUT :-

High-Pass Coefficients (Rectangular Window) ---

-0.0089
0
0.0097
0.0143
0.0106
0
-0.0117
-0.0175
-0.0131
0
0.0148
0.0225
0.0171
0
-0.0202
-0.0314
-0.0247
0

0.0318

0.0524

0.0445

0

-0.0741

-0.1572

-0.2223

0.7407

-0.2223

-0.1572

-0.0741

0

0.0445

0.0524

0.0318

0

-0.0247

-0.0314

-0.0202

0

0.0171

0.0225

0.0148

0

-0.0131

-0.0175

-0.0117

0

0.0106

0.0143

0.0097

0

-0.0089

--- High-Pass Coefficients (Hamming Window) ---

-0.0007

0

0.0009

0.0016

0.0015

0

-0.0024

-0.0044

-0.0039

0

0.0060

0.0103

0.0088

0

-0.0128

-0.0217

-0.0184

0

0.0268

0.0464

0.0410

0

-0.0725

-0.1567

-0.2240

0.7493

-0.2240

-0.1567

-0.0725

0

0.0410

0.0464

0.0268

0

-0.0184

-0.0217

-0.0128

0

0.0088

0.0103

0.0060

0

-0.0039

-0.0044

-0.0024

0

0.0015

0.0016

0.0009

0

-0.0007

--- High-Pass Coefficients (Hanning Window) ---

-0.0000

0

0.0003

0.0008

0.0009

0

-0.0020

-0.0038

-0.0035

0

0.0057

0.0100

0.0087

0

-0.0127

-0.0216

-0.0183

0

0.0267

0.0464

0.0410

0

-0.0726

-0.1568

-0.2243

0.7500

-0.2243

-0.1568

-0.0726

0

0.0410

0.0464

0.0267

0

-0.0183

-0.0216

-0.0127

0

0.0087

0.0100

0.0057

0

-0.0035

-0.0038

-0.0020

0

0.0009

0.0008

0.0003

0

-0.0000

